**ELEC70006**
**EE9-CS1-2**
**EE9-FPN2-06**

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

# DESIGN OF LINEAR MULTIVARIABLE CONTROL SYSTEMS

Tuesday, 6 May 2025, 10:00 AM

Time allowed: 3:00 hours

**There are FOUR questions on this paper.**

**Answer ALL questions.**

*All questions carry equal marks.*

**Any special instructions for invigilators and information for candidates are on page 1.**

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Second Marker(s) : E. Kerrigan

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1. Consider the feedback system in Figure 1, where $G(s)$ is the nominal system model and $K(s)$ is a controller. The transfer matrix $\Delta(s)$ denotes a stable uncertainty. The design specifications are as follows:

   i) The robust stability specification is to design a controller $K(s)$ such that the feedback loop is internally stable when $\|\Delta(j\omega)\| \leq |w(j\omega)|$ for all $\omega$, where $w(s)$ is an appropriate weighting function.

   ii) The performance specifications are to design an internally stabilizing controller such that when $\Delta = 0$, the closed-loop should satisfy:

$$\|u(j\omega)\| < |w_u(j\omega)^{-1}| \|r(j\omega)\|,$$

   for all $\omega$, where $w_u(s)$ is an appropriate weighting function.

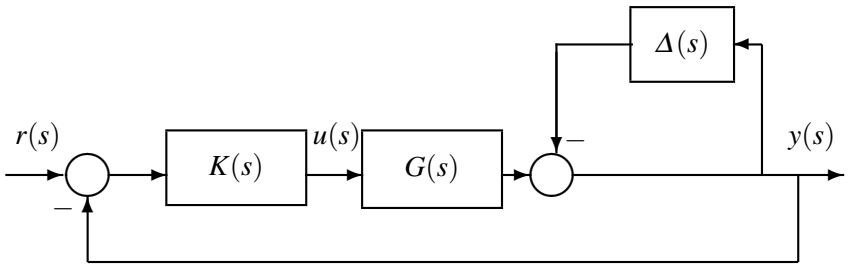



*Figure 1*

   a) State the small gain theorem concerning the internal stability of a feedback loop with a forward transfer matrix $\Delta$ and a feedback transfer matrix $H$. [ 5 ]

   b) Use the small gain theorem to derive $\mathscr{H}_\infty$–norm bounds, in terms of $G(s)$, $K(s)$, $w(s)$ and $w_u(s)$ that are sufficient to achieve the specifications. [ 5 ]

   c) Define appropriate cost, external, control and measured signals and draw a block diagram, showing these signals, the nominal system model, the controller, and weighting functions. [ 5 ]

   d) Derive and draw a generalised regulator problem that captures these sufficient conditions. [ 5 ]

2. a) Consider a state–variable model described by the dynamics

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad y(t) = Cx(t) + Du(t),$$

and let $H(s) = D + C(sI - A)^{-1}B$ denote the corresponding transfer matrix.

i) Let $\gamma > 0$ be given. Derive sufficient conditions in the form of LMIs for the stability of $A$ and for $\|H\|_\infty < \gamma$. [ 5 ]

ii) Derive a dual set of conditions for $H(s)$ stable and $\|H\|_\infty < \gamma$. [ 5 ]

b) Consider the output injection problem in Figure 2. Let $T_{yr}(s)$ denote the transfer matrix from $r$ to $y$. An internally stabilizing gain matrix $L$ is to be designed such that, for given $\gamma > 0$, $\|T_{yr}\|_\infty < \gamma$.

i) Derive a state space realization for $T_{yr}(s)$. [ 5 ]

ii) Derive sufficient conditions for the existence of a feasible $L$. Your conditions should be in the form of the existence of solutions to linear matrix inequalities. [ 5 ]

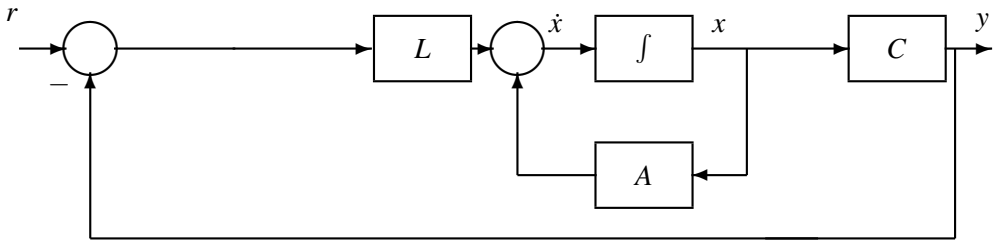



*Figure 2*

3. Consider the regulator in Figure 3 for which it is assumed that $B$ has full column rank, the pair $(A,B)$ is controllable and $x(0)=x_0$. A stabilizing state-feedback gain matrix $F$ is to be designed such that the cost function

$$J=\left\|\begin{bmatrix} x \\ u \end{bmatrix}\right\|_2^2$$

is minimized.

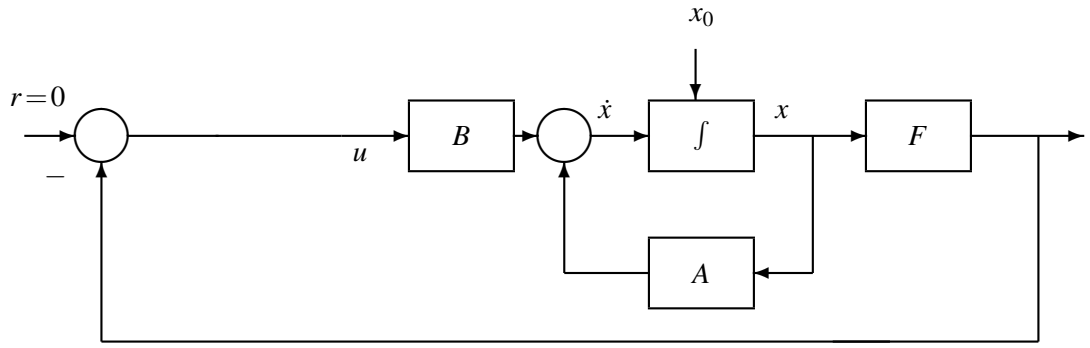



*Figure 3*

a) Define a suitable Lyapunov function $f(x(t))$ and evaluate an expression for the integral

$$\int_0^\infty \dot{f}(x(t))dt.$$

[ 5 ]

b) By defining a suitable cost function, using the answer to Part (a) and completing a square, find an expression for $F$ that achieves the design specifications in terms of the solution of an algebraic Riccati equation. Give the minimum value of the cost function as well as the algebraic Riccati equation. [ 5 ]

c) Prove that, for $F$ chosen in Part (a), the closed-loop system in Figure 3 is stable. State clearly the assumptions required to guarantee stability. [ 5 ]

d) Prove that, for $F$ chosen in Part (a), the loop gain in Figure 3 is minimum-phase. [ 5 ]

4. Consider the regulator shown in Figure 4 for which it is assumed that the pair $(A,B_2)$ is controllable and the triple $(A,B_1,C_1)$ is minimal. Let $z = [\ z_1^T \ \ z_2^T\ ]^T$. It is required to regulate $z$ in response to the external finite energy disturbance $w$ and the initial state $x(0) = x_0$.

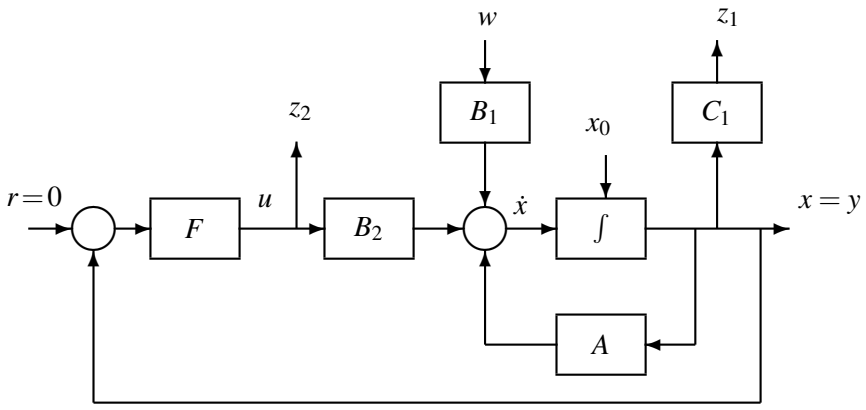



*Figure 4*

a) For this part, take $w = 0$. A stabilizing state-feedback gain matrix $F$ is to be designed such that $J_0 := \|z\|_2^2$ is minimized. By defining a suitable Lyapunov function in terms of an auxilliary variable $P$, find an expression for $F$ that minimizes $J_0$. Give also the algebraic Riccati equation that needs to be satisfied. Prove that the closed-loop system is stable. [ 5 ]

b) For this part, take $x_0 = 0$. Let $H(s)$ denote the transfer matrix from $w$ to $z$ and let $\gamma > 0$ be given. A stabilizing state-feedback gain matrix $F$ is to be designed such that $\|H\|_\infty < \gamma$ for a given $\gamma > 0$.

i) By defining a Lyapunov function and a cost function $J$ involving an auxiliary variable $X$, and carrying out two completions of squares, derive an expression for $J$ that can be used to solve the problem. [ 5 ]

ii) Use the expression for $J$ to formulate the design problem and derive sufficient conditions for its solution. Your conditions should be in the form of the existence of a solution to an algebraic Riccati equation. It should also include an expression for $F$ and an expression for the worst–case disturbance $w^*$. [ 5 ]

c) For this part, we assume $w \neq 0$ and $x_0 \neq 0$. By comparing the solutions to part (a) and (b) above, suggest a procedure for a tradeoff between performance and robustness. [ 5 ]